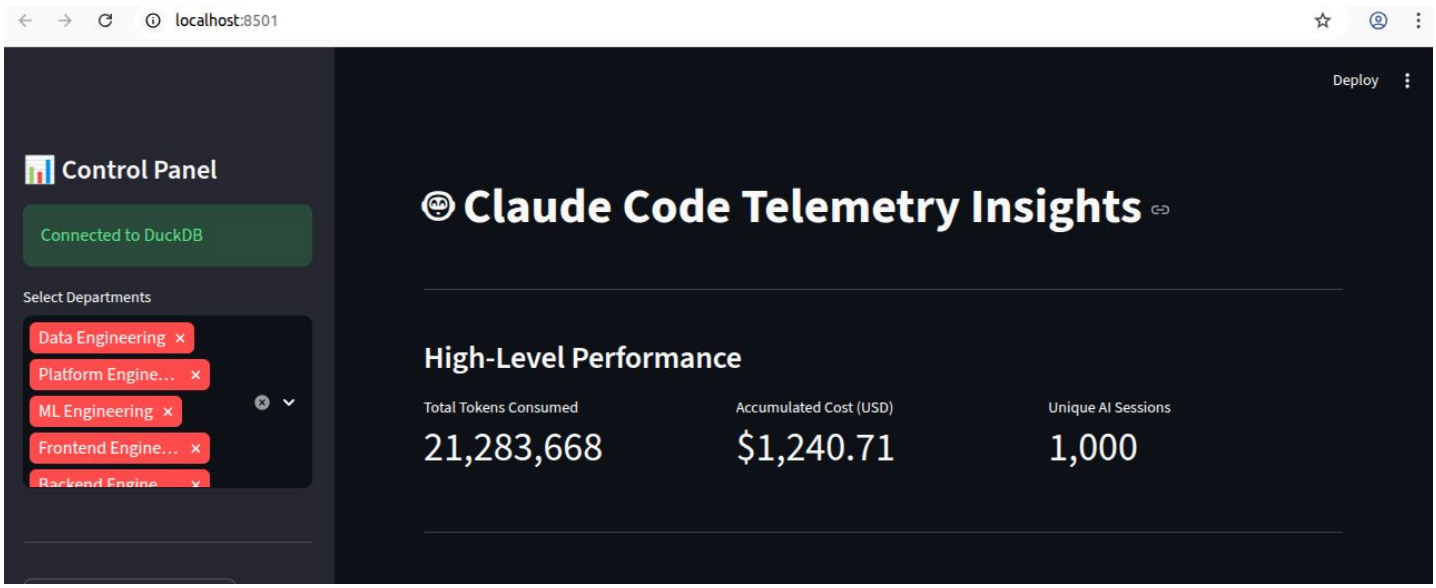
Executive Overview & Technical Stack

Headline: Claude Code Telemetry Analytics — Data-Driven Insights

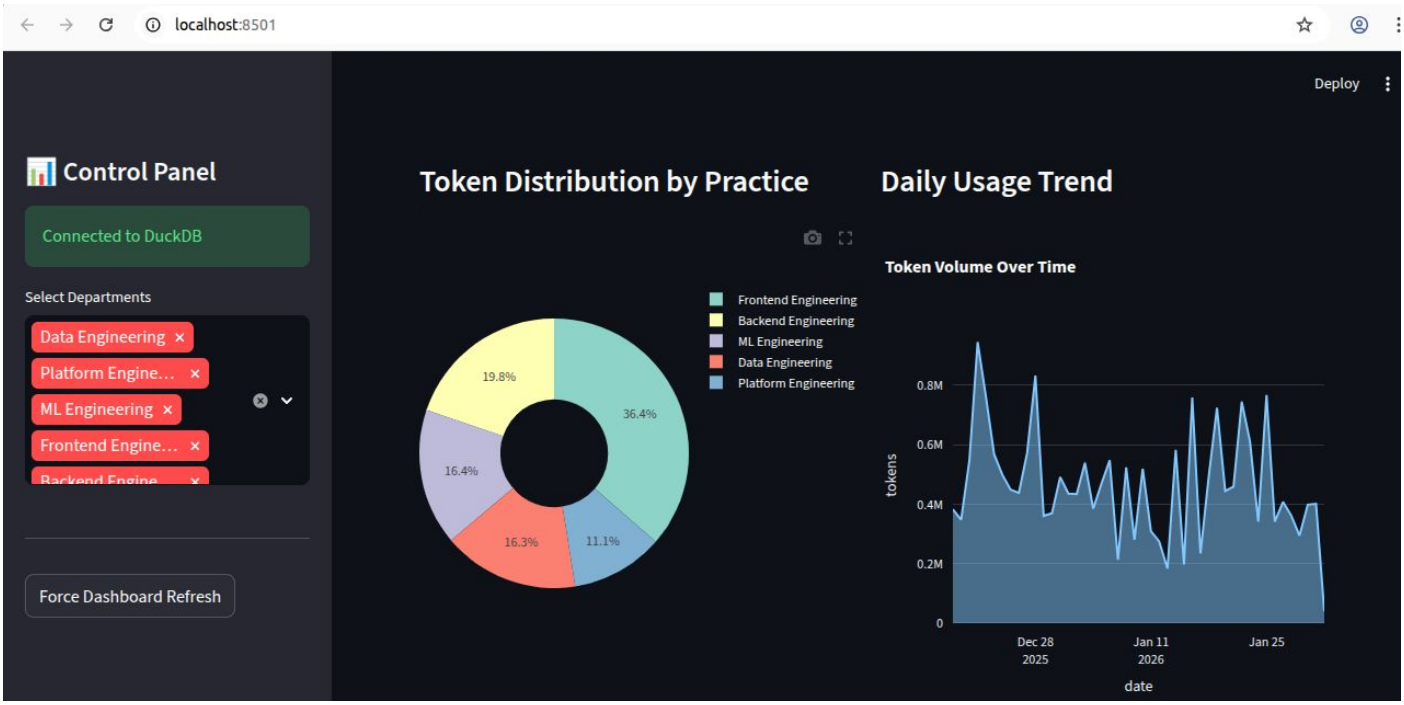
- **Project Goal:** To transform raw developer telemetry into actionable business intelligence regarding AI adoption and costs.
- **Technical Stack:**
  - **Engine:** DuckDB (OLAP-optimized for high-speed analytical queries).
  - **Frontend:** Streamlit with Plotly interactive visualizations.
  - **Advanced Features:** Linear Regression for forecasting and Z-Score analysis for anomaly detection.
- **High-Level KPI (from Screenshot 1):**
  - **21.2M+** Tokens Consumed.
  - **$1,240.71** Total Invested in API costs.
  - **1,000** Unique AI Sessions analyzed.



User Behavior & Departmental Impact

Headline: Token Distribution by Engineering Practice

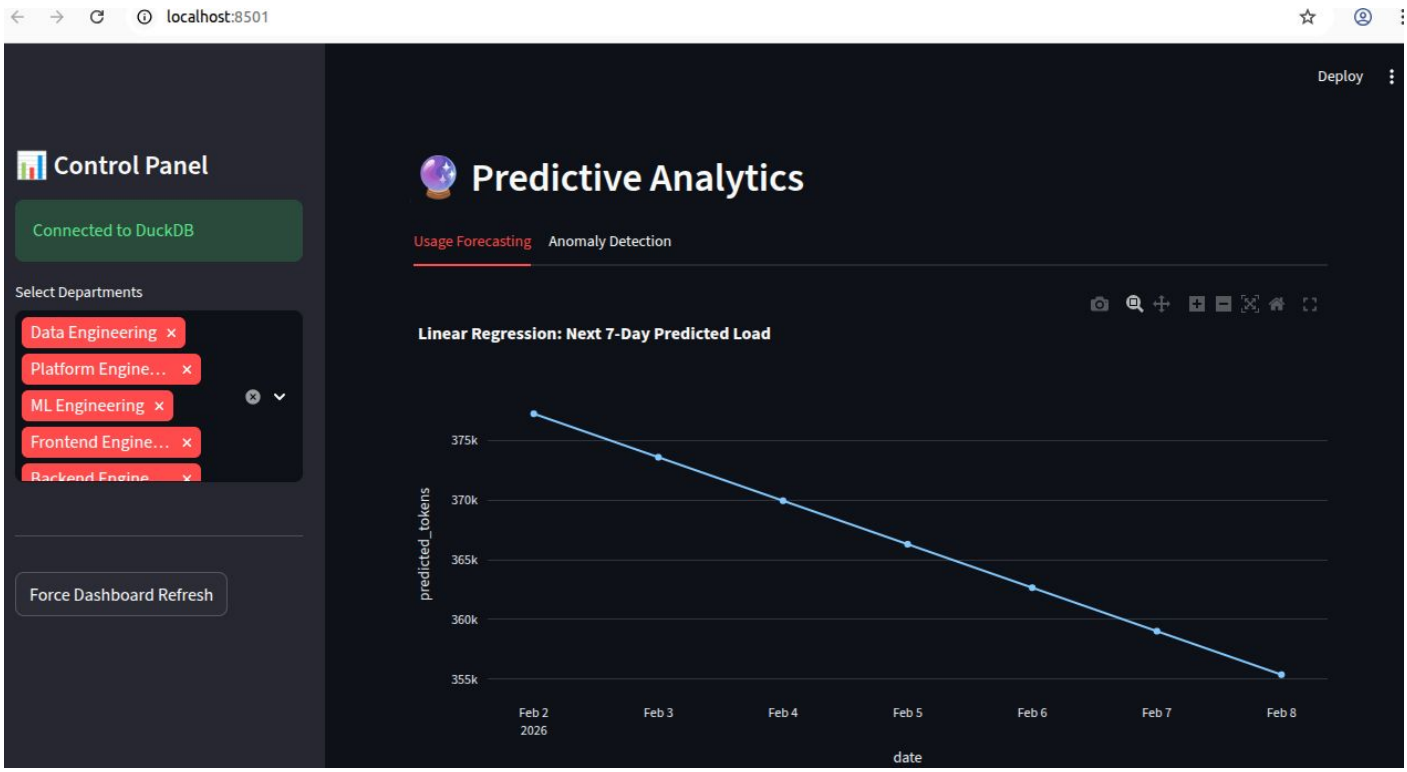
- **Insight (The "Who"): Frontend Engineering** is the dominant user group, accounting for **36.4%** of total token volume.
- **Story:** This indicates that Claude Code is most heavily integrated into UI/UX development workflows.
- **Usage Trends (from Screenshot 2):**
  - **Backend & ML Engineering** follow at **19.8%** and **16.4%** respectively.
  - **Daily Usage Trend:** Shows high volatility with periodic spikes (up to 0.8M tokens/day), suggesting Claude is used for intensive "deep work" sessions rather than just minor queries.



Predictive Analytics & Forecasting

Headline: Predictive Load Management (Bonus Feature)

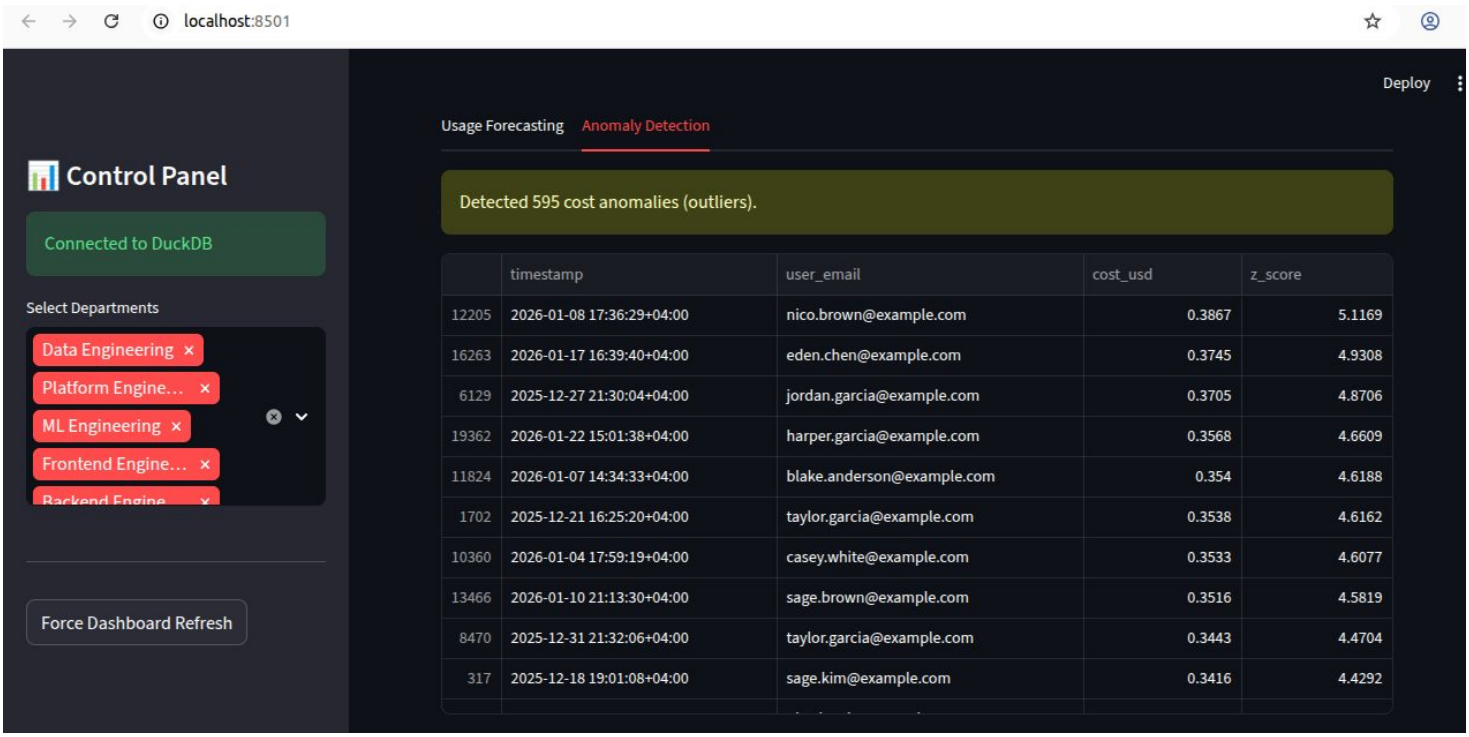
- **Mechanism:** Implemented a **Linear Regression** model to project next 7-day resource requirements.
- **Finding (from Screenshot 3):**
  - The model predicts a gentle downward slope in token demand (from ~378k to ~355k).
  - **Strategic Value:** This allows management to anticipate API billing cycles and ensure that token quotas are aligned with actual developer velocity.



Anomaly Detection & Cost Control

Headline: Automated Anomaly Detection via Z-Scores

- **Feature:** Implemented an outlier detection system using Statistical Z-Scores.
- **Finding (from Screenshot 4):**
  - **595 cost anomalies** detected.
  - **Example Outlier:** A single request costing ~$0.38 with a **Z-Score of 5.11**.
- **Business Value:** This system prevents "billing shock" by identifying inefficient prompt behaviors or accidental automated loops that consume excessive resources.



	timestamp	user_email	cost_usd	z_score
12205	2026-01-08 17:36:29+04:00	nico.brown@example.com	0.3867	5.1169
16263	2026-01-17 16:39:40+04:00	eden.chen@example.com	0.3745	4.9308
6129	2025-12-27 21:30:04+04:00	jordan.garcia@example.com	0.3705	4.8706
19362	2026-01-22 15:01:38+04:00	harper.garcia@example.com	0.3568	4.6609
11824	2026-01-07 14:34:33+04:00	blake.anderson@example.com	0.354	4.6188
1702	2025-12-21 16:25:20+04:00	taylor.garcia@example.com	0.3538	4.6162
10360	2026-01-04 17:59:19+04:00	casey.white@example.com	0.3533	4.6077
13466	2026-01-10 21:13:30+04:00	sage.brown@example.com	0.3516	4.5819
8470	2025-12-31 21:32:06+04:00	taylor.garcia@example.com	0.3443	4.4704
317	2025-12-18 19:01:08+04:00	sage.kim@example.com	0.3416	4.4292

Conclusion & Future Roadmap

Headline: Final Findings & Recommendations

- **Summary:** Claude Code has become a critical tool for the Frontend and Backend teams, with a highly positive ROI based on session volume.
- **Key Recommendations:**
  1. **Optimize High-Cost Prompts:** Investigate the identified anomalies (Z-Score > 3) to retrain users on prompt efficiency.
  2. **Scale Resources:** Maintain current funding levels as usage remains high and consistent across the most productive engineering practices.
  3. **Refine Forecasting:** Incorporate seasonality (sprint cycles) into the ML model for even higher accuracy.